

Todd H. Albert, Ph.D.

Engineering Executive who rebuilds complex software, teams, and systems

30+ years Software • 15+ years Leadership (iD) • 20+ years Academia

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Executive Profile

Engineering executive with 30+ years of technical depth and 15+ years leading teams through complex rebuilds. I specialize in taking software platforms, engineering organizations, and delivery systems that have become fragile, stalled, or overly complex — then turning them into scalable, understandable, high-performing systems. Former NASA-funded scientist, educator, founder, and hands-on architect with executive experience across SaaS, EdTech, AdTech, AI, cloud infrastructure, and private-equity-backed transformation.

Signature Strengths

- Rebuilding complex engineering systems without losing organizational trust
- Translating technical ambiguity into executive-level decisions
- Modernizing legacy platforms, delivery systems, and cloud operations
- Leading distributed engineering, DevOps, QA, and integration teams
- Connecting scientific rigor, product judgment, and hands-on technical depth
- Building AI-enabled products that solve real user and operational problems

Professional Experience

VP of Engineering, CYPHER Learning

Global SaaS learning platform | Private-equity-backed | Remote

Oct 2024 – present

- Lead engineering, DevOps, QA, and integrations for a global learning platform serving millions of users across multiple production environments.
- Rebuilt engineering operating rhythms across planning, delivery, QA, cloud operations, and executive reporting, moving the organization toward clearer accountability and more predictable execution.
- Reduced cloud infrastructure spend by over \$1.2M annually while improving platform resilience, performance, and operational visibility.
- Manage a 32+ person distributed organization and \$4.4M budget across engineering, DevOps, QA, and integrations.
- Present technology strategy, roadmap progress, risk, and financial metrics to private-equity board stakeholders.
- Drive modernization across AI, automation, security, compliance, and data visibility, including SOC 2, GDPR, and EAA-aligned initiatives.

VP of Application Development, Site Impact

AdTech platform modernization | Multi-region cloud architecture | Hybrid

Jan 2024 – Sep 2024

- Rescued a stalled \$4.3M platform initiative after 2.5 years of missed deadlines by simplifying the architecture, reducing technical ambiguity, and resetting team execution practices.
- Replatformed core AdTech systems for multi-region resilience, reducing projected infrastructure spend by approximately \$600K annually.
- Built and led engineering, DevOps, and SRE practices from the ground up, including CI/CD, observability, incident readiness, and infrastructure-as-code.
- Managed global engineering delivery, vendor relationships, executive reporting, and a \$3.6M budget.
- Partnered with senior leadership on quarterly business planning, board materials, operating metrics, and team performance systems.

CTO, Edgy Nerd

Founder-led product studio | AI, SaaS, fintech, and cloud platforms | Boca Raton

Jan 2020 – Feb 2024

- Founded and led a product engineering studio delivering SaaS, fintech, AI, and cloud platforms for clients across regulated and high-growth markets.
- Architected and launched AI-powered applications, real-time payment systems, analytics platforms, and cloud-native products across AWS, Azure, and GCP.
- Built resilient, secure, and compliant systems for clients with uptime, data sensitivity, and operational reliability requirements.
- Engineered and scaled a nationally recognized AI diarization application, combining speech processing, automation, and production-grade cloud architecture.
- Managed product strategy, architecture, delivery, client relationships, engineering teams, vendors, budgets, and P&L across multiple concurrent engagements.

Head Software Engineer, MyPhoto / Polaroid Photo Bar

E-commerce and production automation platform | Computer vision and custom fulfillment systems | Delray Beach

Apr 2017 – Sep 2020

- Rebuilt the company's custom e-commerce and fulfillment platform in under six months, modernizing from PHP to Node.js, React, and cloud-native infrastructure.
- Led architecture, data engineering, application development, security, CI/CD, and deployment for a high-volume personalized photo-commerce platform.
- Migrated the platform to a multi-cloud architecture, improving performance while increasing conversion rate from 2% to 10% and cutting bounce rate by 50%.
- Automated production workflows with AI, computer vision, Python, React Native, and Electron, reducing production time by 66% and increasing productivity 10x.
- Invented an "instant upload" experience that removed customer friction and improved the speed, reliability, and usability of the ordering flow.

Lead Software Engineer, MadValorem

Real estate data platform rebuild | SEO-scale content automation and mapping systems | Delray Beach

Aug 2013 – Apr 2015

- Rebuilt a stalled \$1.2M real estate data platform from the ground up, rapidly migrating the core system from .NET to a LAMP architecture.
- Solved a critical mapping and geospatial search problem that had blocked a 12-engineer team and remained unresolved in public developer forums.
- Built a deterministic content-generation system that produced structured descriptions for 600,000+ properties, developments, neighborhoods, cities, counties, and states.
- Developed a scalable SEO architecture that ranked property-related professionals in the top 10 Google results across Florida city and service-category combinations.
- Partnered directly with the CEO on product strategy, platform architecture, and business-critical technical decisions, earning a 33% equity stake in the company.

CTO / Founder / Advisory Work

TapIn | Hemp Temps | Nebular Agency | others

2015 – 2026

- Founded and led technical delivery across SaaS, fintech, AI, AR, e-commerce, and education platforms.
- Built and managed engineering teams, vendor relationships, client delivery, budgets, and product architecture.
- Delivered platforms for clients and partners including PBS Kids, AccuWeather, Ally Bank, Sprint, and early-stage SaaS ventures.
- Developed AI, AR, and automation systems generating measurable revenue, productivity, and operational gains.

Scientific and Academic Foundation

- Ph.D., Florida State University; M.S., The Ohio State University; B.S., University of Florida.
- NASA Graduate Research Fellow; NASA
- NASA Mission to Planet Earth Scientific Grant Evaluator
- Founded and led the first Meteorology BA/BS program in Tennessee.
- Taught 2,500+ students and helped grow a university major by 7x, preserving department viability.
- Published peer-reviewed research on glaciers, remote sensing, and climate systems.

Writing, Research, and Thought Leadership

Albert, T.H., 2026, Your Customers Don't Want Your AI Assistant, <https://impactvelocity.studio/notes/your-customers-dont-want-your-ai-assistant>

Albert, T.H., 2026, Out of the Fish Tank, <https://www.amazon.com/dp/B0H4766WZ9>

Albert, T.H., 2025, The least-used but most-powerful debugging tool for engineers, <https://toddhalbert.medium.com/the-least-used-but-most-powerful-debugging-tool-for-engineers-77aec7cda226>

Albert, T.H., 2024, 7 Reasons Why Most Products Fail, <https://toddhalbert.medium.com/7-reasons-why-most-products-fail-ef5ed8c17db8>

Albert, T.H., 2021, Deploying an Express API to Google App Engine, <https://medium.com/boca-code/deploying-an-express-api-to-google-app-engine-ac7a188c6608>

Albert, T.H., 2021, Deploying an Express API on Firebase Hosting, <https://medium.com/boca-code/the-basic-process-is-that-we-will-use-firebase-cloud-functions-to-create-a-single-function-app-13ba3b852077>

Albert, T.H., 2021, More jobs than coders — Boca Code rises to the challenge, <https://toddhalbert.medium.com/more-jobs-than-coders-boca-code-rises-to-the-challenge-fc75e1b47802>

Albert, T.H., 2020, Why I Switched from Kubernetes to Firebase Hosting, <https://toddhalbert.medium.com/why-i-switched-from-kubernetes-to-firebase-hosting-ae1fdac62439>

Albert, T.H., 2016, C. Huggel, A. Racoviteanu, Y. Arnaud, W. Silvero, and A. Klein, Northern Andes, in *Global Land Ice Measurement from Space*, edited by J. Kargel

Albert, T.H., 2010, Glaciers: Continental, in *Encyclopedia of Geography*, edited by B. L. Warf, Sage Publishing. **Winner of "Outstanding Reference Source" award by the American Library Association**

Albert, T.H., 2002, Evaluation of remote sensing techniques for ice-area classification applied to the tropical Quelccaya Ice Cap, Peru, *Polar Geography*, 26 (3), 210-226

Albert, T.H., 2008, Assessment of glacier mass balances from small tropical glaciers to the ice sheet of Greenland, VDM Verlag Dr. Mueller e.K., Saarbrücken, Germany, 160 pp

Technical Scope

Leadership: Engineering strategy, org design, platform modernization, board reporting, distributed teams, budgets, delivery systems

Platforms: SaaS, EdTech, AdTech, fintech, e-commerce, AI-enabled products

Architecture: Cloud platforms, Rails/Node systems, APIs, data platforms, security, observability, CI/CD

Cloud & Operations: AWS, GCP, Azure, Kubernetes, Terraform, DevOps, SRE, cost optimization

AI & Data: LLM-enabled workflows, automation, analytics, data pipelines, computer vision